[LAB Report - 1]

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Course code : CSE 260

Section : 01

Group : 01

Experiment Name:

Experiment # 1: Familiarization with Fundaemental Logic Gates

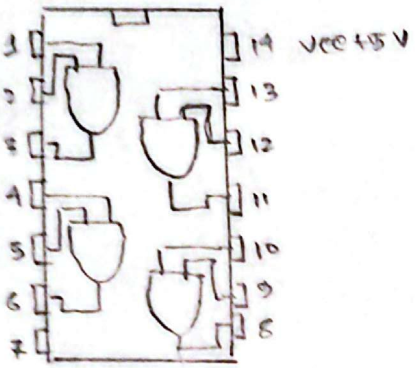
Experiment # 2: Universal Gates, Applications of Boolean Algebra.

Experiment # 3: Parity Bit checker and Generator

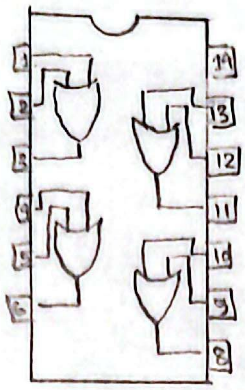
Required components for Experiment - 1,2,3:

1. IC 7408 x 1
2. IC 7432 x 1
3. IC 7404 x 1
4. IC 7400 x 1
5. IC 7402 x 1
6. IC 7486 x 1
7. IC 4077 x 1

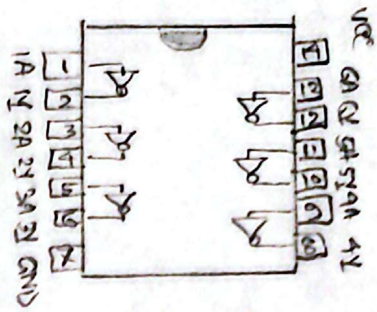
[Experiment-1] Pin diagrams of ICs [Setup]



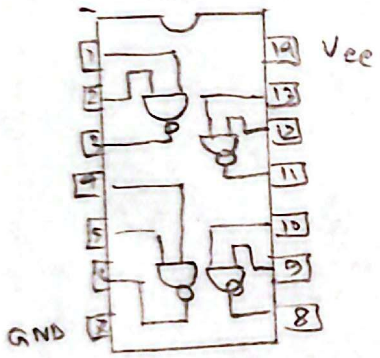
pin layout of 7408



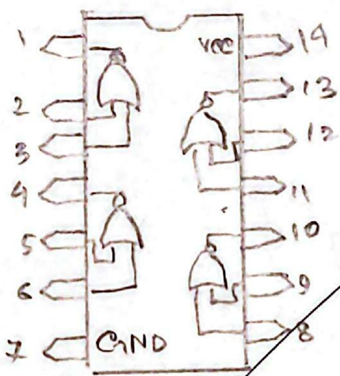
Pin layout of 7432



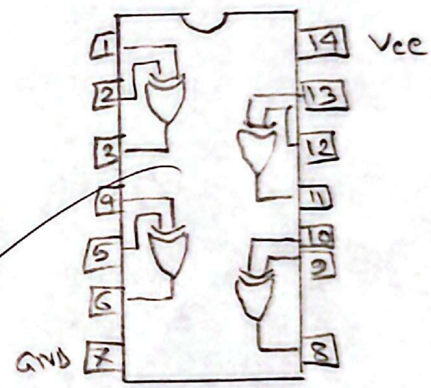
pin layout of 7404



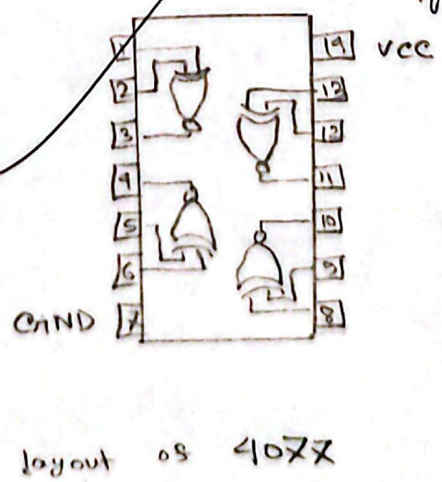
pin layout of 7400



pin layout of 7402



pin layout of 7486



pin layout of 4077

[Experiment - 2]

[setup]

Name of the experiment: Universal Gates, Application of Boolean Algebra.

Objective:

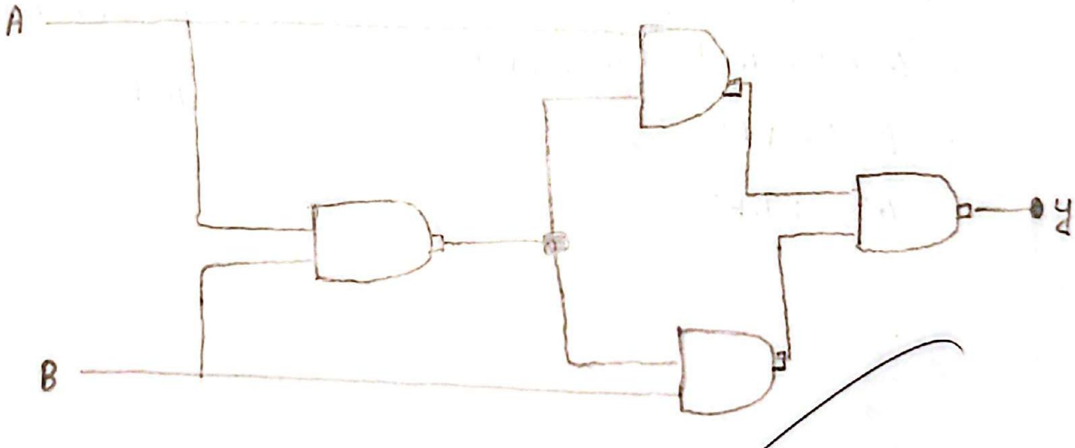
(i) To investigate the rules of Boolean algebra.

(ii) To gain experince working with practical circuits.

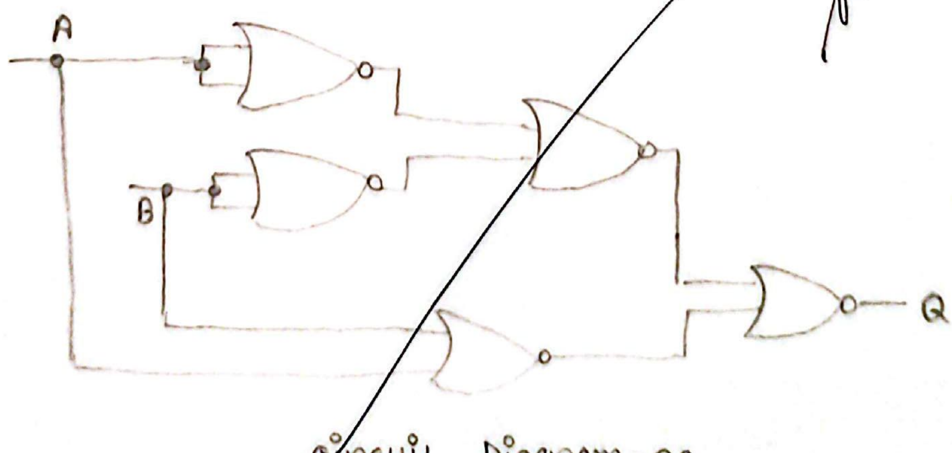
(iii) To simplify a complex function using Boolean algebra.

Required components: IC 7400, IC 7402

Experimental setup:



circuit Diagram-01



circuit Diagram-02

[Experiment - 3]

[Setup]

Name of the experiment:

Parity Generator and Checker

Objective:

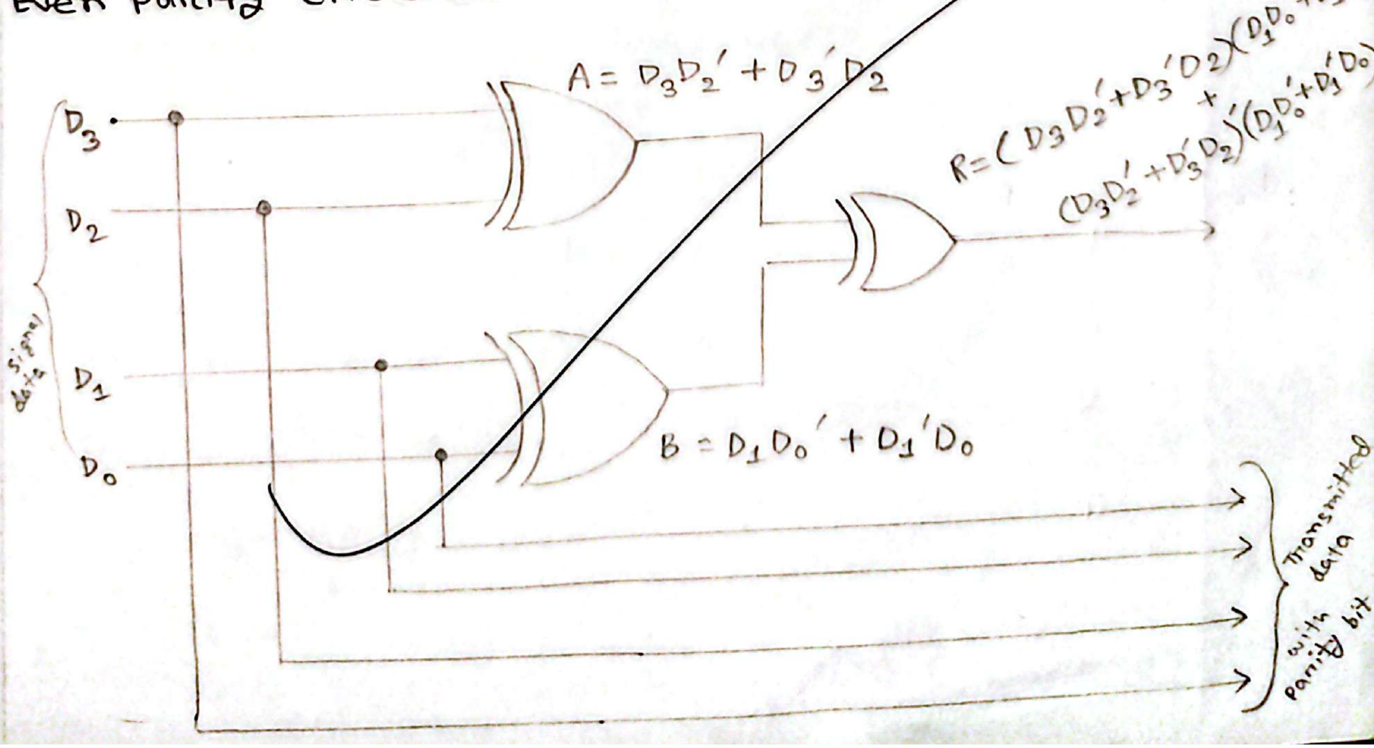
To design and implement an Even Parity generator and Even parity checker using XOR gates (IC-7486).

Required components and Equipments:

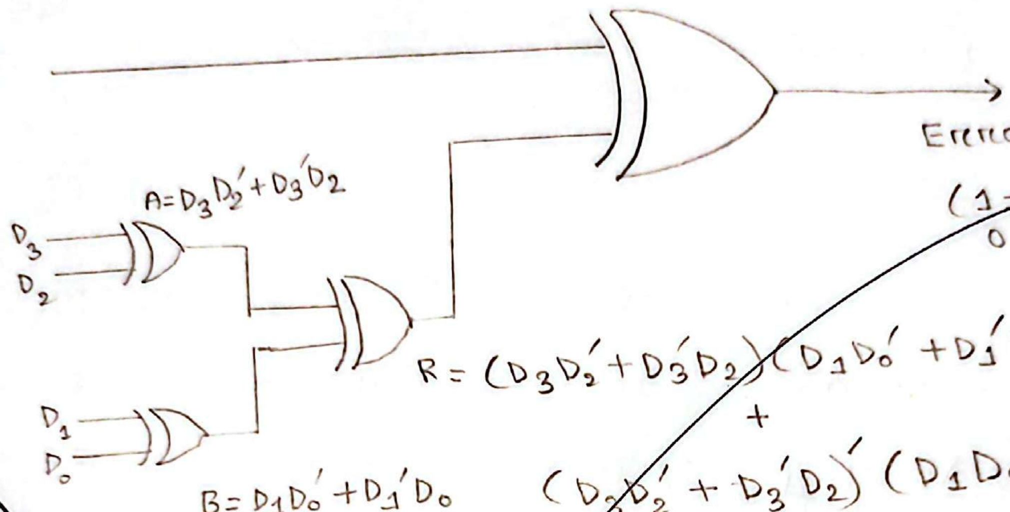
1. AT-700 portable analog / Digital Labratory
2. 7400 × 3

Experimental setup:

Even parity generator ~~checker~~



Even parity checker



Error(E) $= PR' + P'R$
(1 = error
0 = no error)

$R = (D_3 D_2' + D_3' D_2)(D_1 D_0' + D_1' D_0)'$
$+$
$(D_3 D_2' + D_3' D_2)'(D_1 D_0' + D_1' D_0)$

[Experiment_3 Result]

Results: Even parity generator's output

D_3	D_2	D_1	D_0	Parity
1	0	0	1	0
0	0	0	1	1
1	1	1	1	0
0	0	0	0	0

Even parity checker's output

P	D_3	D_2	D_1	D_0	Error
1	1	0	0	1	1
0	0	0	0	1	1
0	1	1	1	1	0
1	0	0	0	0	1

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[Result]

[Experiment-1]

Logic gate symbols with corresponding truth table:

~~Not~~ NOT gate ( )

A	B
0	1
1	0

AND gate ( )

A	B	Output
0	0	0
1	0	0
0	1	0
1	1	1

OR gate ( )

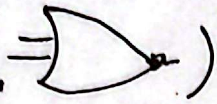
A	B	Output
0	0	0
1	0	1
0	1	1
1	1	1

XOR gate ( )

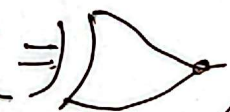
A	B	Output
0	0	0
1	0	1
0	1	1
1	1	0

NAND gate ( )

A	B	Output
0	0	1
1	0	1
0	1	1
1	1	0

NOR ()

A	B	Output
0	0	1
1	0	0
0	1	0
1	1	0

XNOR ()

A	B	Output
0	0	1
1	0	0
0	1	0
1	1	1

[Experiment - 2]
[Result]

Results :

for circuit - 1,

A	B	output
0	0	0
0	1	1
1	0	1
1	1	0

$(((A \cdot (AB)')' \cdot (B \cdot (AB)')')'$

$= (A \cdot (AB)') + (B (AB)')$

$= A \cdot (A' + B') + (B (A' + B'))$

$= A \cdot A' + AB' + BA' + BB'$

$= AB' + BA'$

$= A'B + AB'$

for circuit - 2

A	B	output
0	0	0
0	1	1
1	0	1
1	1	0

$((AB) + (A+B)')'$

$= (A \cdot B)' \cdot (A+B)$

$= (A' + B') \cdot (B + A)$

$= A'B + A'A + B'B + AB'$

$= A'B + AB'$

[Experiment - 3]
[Result]

Results: Even parity generator's output

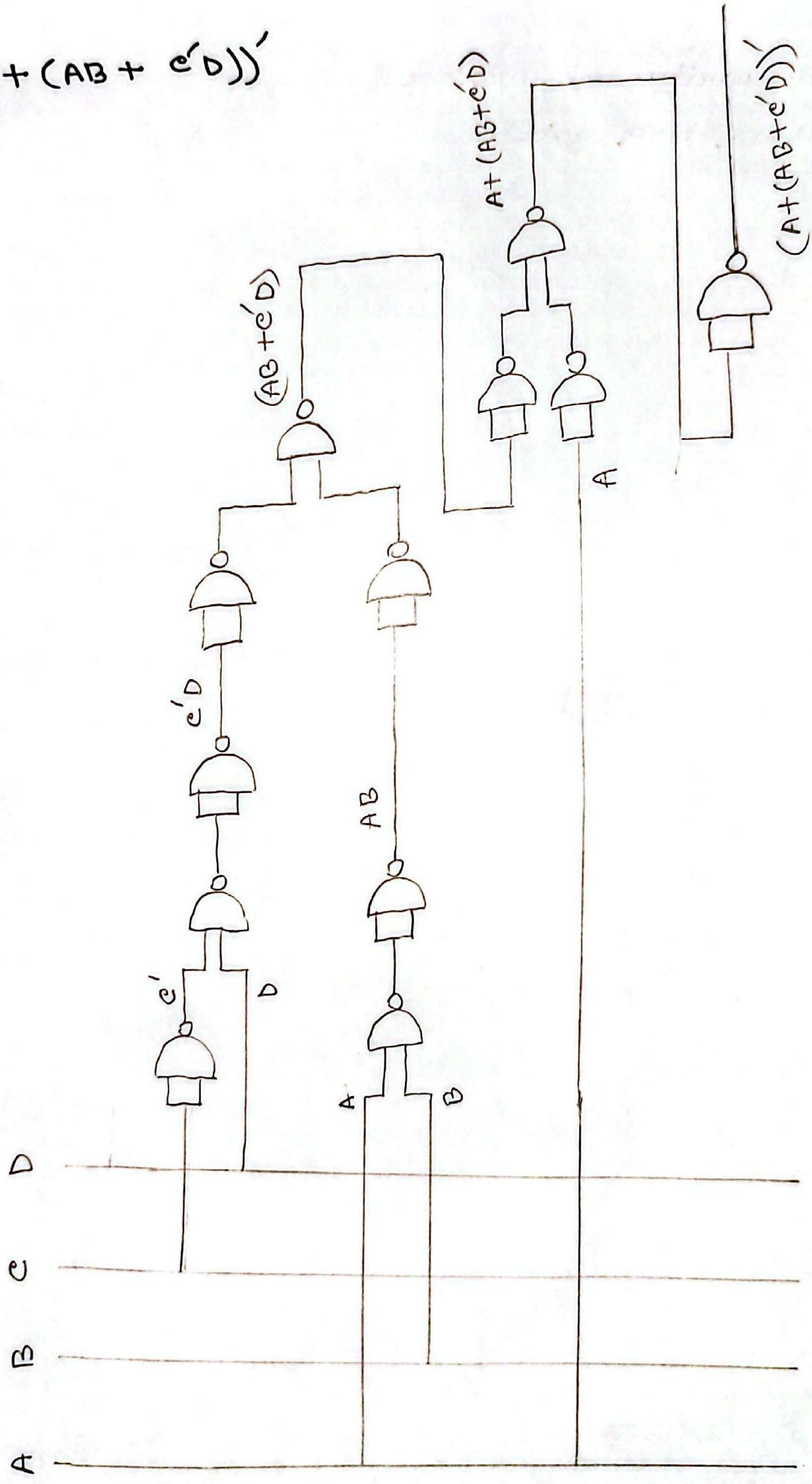
D_3	D_2	D_1	D_0	Parity
1	0	0	1	0
0	0	0	1	1
1	1	1	1	0
0	0	0	0	0

Even parity checker's output

P	D_3	D_2	D_1	D_0	Error
1	1	0	0	1	1
0	0	0	0	1	1
0	1	1	1	1	0
1	0	0	0	0	1

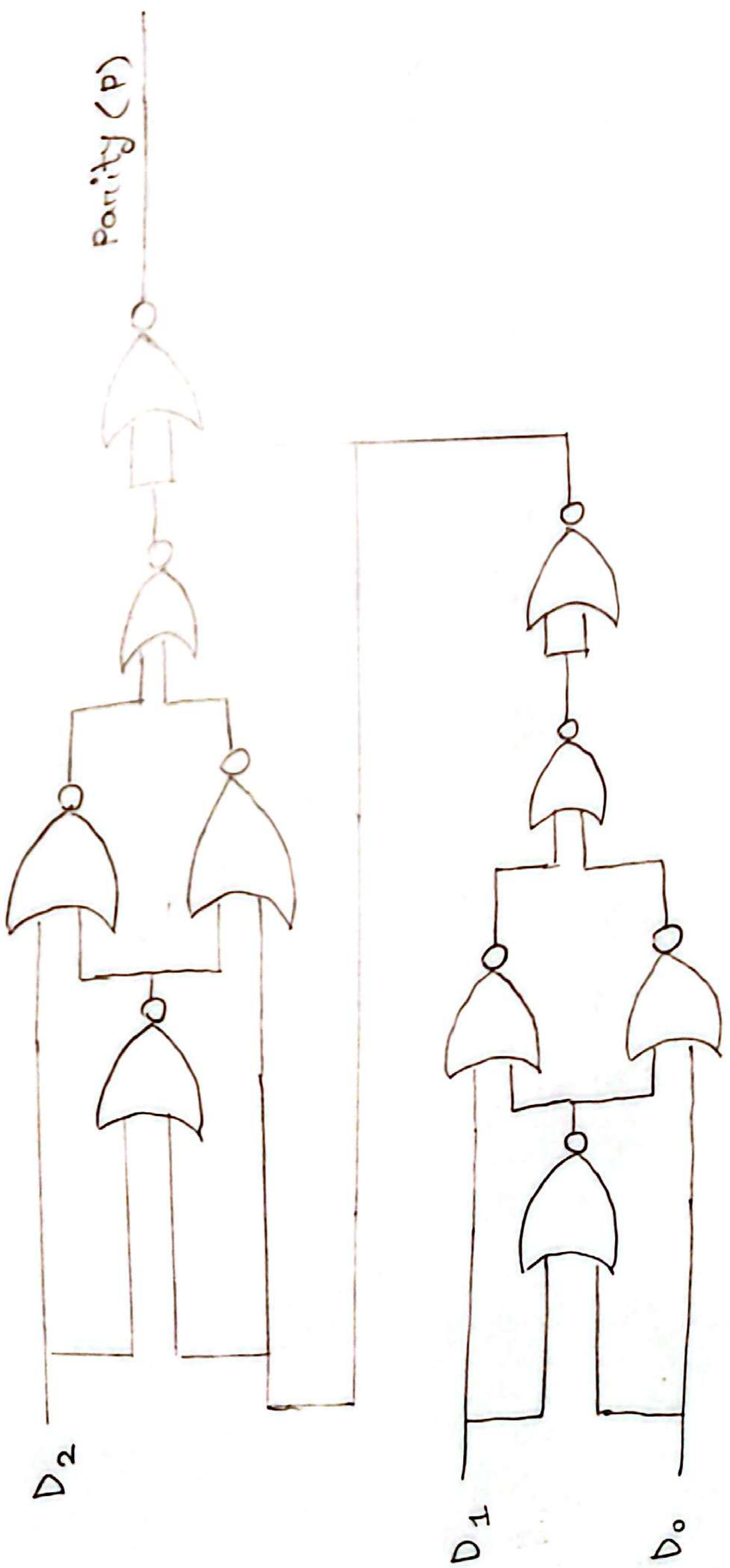
Answer to the question no 1

$(A+(AB+C'D))'$



Answer to the question no 2

3-bit parity generator :
using NOR gates



3 bit parity checker:

using NOR Gates